

Q1. First Law of Thermodynamics.

The **First Law of Thermodynamics** states that:

"Energy can neither be created nor destroyed. It can only be transformed from one form to another."

When heat is supplied to a system, it is used to:

- Increase the **internal energy** of the system.
- Perform **work** on the surroundings.

Mathematical Expression

If

- Q = Heat supplied to the system
- W = Work done by the system
- ΔU = Change in internal energy

Then,

$$Q = \Delta U + W$$

or

$$\Delta U = Q - W$$

This is the **mathematical form of the First Law of Thermodynamics**.

Applications of the First Law

1. Isochoric (Constant Volume) Process

Since volume remains constant,

$$W = 0$$

Therefore,

$$Q = \Delta U$$

Conclusion: All supplied heat increases internal energy.

2. Isothermal (Constant Temperature) Process

Temperature remains constant.

Hence,

$$\Delta U = 0$$

Therefore,

$$Q = W$$

Conclusion: All supplied heat is converted into work.

3. Adiabatic Process

No heat enters or leaves the system.

$$Q = 0$$

Therefore,

$$\Delta U = -W$$

Conclusion: Work is done at the expense of internal energy.

4. Cyclic Process

After one complete cycle,

$$\Delta U = 0$$

Therefore,

$$Q = W$$

Conclusion: Net heat supplied equals net work done.

Diagram

Limitations

1. It does **not indicate the direction** of heat flow.
2. It does **not explain why heat flows from hot to cold**.
3. It **cannot determine whether a process is reversible or irreversible**.
4. It **does not provide the maximum possible efficiency** of heat engines.

Importance

- Forms the basis of thermodynamics.
- Used in studying heat engines, refrigerators, and air conditioners.
- Explains the principle of conservation of energy in thermal processes.
- Widely applied in physics and engineering.

Q2. Second Law of Thermodynamics.

The **Second Law of Thermodynamics** states that:

"Heat cannot flow spontaneously from a colder body to a hotter body without external work."

It explains the **direction of heat flow** and shows that not all heat can be converted into useful work.

Kelvin–Planck Statement

According to Kelvin–Planck:

"It is impossible to construct a heat engine that converts all the heat absorbed from a source completely into work without rejecting some heat to a sink."

Explanation

- A heat engine absorbs heat Q_1 from a hot reservoir.
- It converts only part of this heat into work W .
- The remaining heat Q_2 is rejected to a cold reservoir.

Therefore,

$$Q_1 = W + Q_2$$

Since $Q_2 \neq 0$, **100% efficiency is impossible**.

Clausius Statement

According to Clausius:

"Heat cannot, by itself, flow from a colder body to a hotter body."

Explanation

- Heat naturally flows from **hot → cold**.
- To transfer heat from **cold → hot**, external work is required.

Example: Refrigerator and Air Conditioner.

Mathematical Representation

For a heat engine,

$$Q_1 = W + Q_2$$

where:

- Q_1 = Heat absorbed from the hot reservoir
- Q_2 = Heat rejected to the cold reservoir
- W = Work done by the engine

Efficiency of the engine:

$$\eta = \frac{W}{Q_1}$$

Since $Q_2 > 0$,

$$\eta < 1$$

Thus, **no heat engine can have 100% efficiency**.

Diagram

Importance of the Second Law

1. Explains the **natural direction of heat flow**.
2. Shows that **complete conversion of heat into work is impossible**.
3. Forms the basis for the design of **heat engines**.
4. Explains the working of **refrigerators and air conditioners**.
5. Introduces the concept of **entropy**, which measures the disorder or randomness of a system.

Applications

1. Heat Engines

Steam engine, petrol engine , diesel engine

Only a part of heat is converted into work.

2. Refrigerators

Heat is transferred from a cold region to a hot region by supplying external work.

3. Air Conditioners

Operate on the same principle as refrigerators by removing heat from indoor air and releasing it outside.

Limitations

1. It does **not describe the amount of heat transferred**.
2. It does **not provide the exact efficiency** of all engines.
3. It **cannot replace the First Law**, which deals with conservation of energy.

Q3. Define Entropy in thermodynamics.

Entropy is a thermodynamic property that measures the **degree of disorder, randomness, or unavailability of energy** in a system.

It is represented by the symbol S .

Simple Definition (for exams):

Entropy is the measure of randomness or disorder of a thermodynamic system.

Mathematical Expression

For a reversible process,

$$dS = \frac{\delta Q_{rev}}{T}$$

or, for a finite change,

$$\Delta S = \frac{Q_{rev}}{T}$$

where,

- S = Entropy
- Q_{rev} = Heat absorbed reversibly
- T = Absolute temperature (Kelvin)

SI Unit

$$\text{Joule per Kelvin (J/K)}$$

Change in Entropy

1. Reversible Process

$$\Delta S = \frac{Q_{rev}}{T}$$

Entropy changes gradually.

2. Irreversible Process

Entropy of the universe **increases**.

3. Adiabatic Reversible Process

Since,

$$Q = 0$$

Therefore,

$$\Delta S = 0$$

Entropy remains constant (an **isentropic process**).

Properties of Entropy

1. Entropy is a **state function**.
2. It depends only on the **initial and final states** of the system.
3. Its SI unit is **J/K**.
4. Entropy increases with increasing temperature.
5. Entropy increases when a solid melts or a liquid evaporates.
6. In an isolated system, entropy never decreases.

Importance of Entropy

1. Determines the direction of spontaneous processes.
2. Explains why heat flows naturally from hot to cold.
3. Helps calculate the efficiency of heat engines.
4. Indicates the degree of disorder in a system.
5. Plays a key role in thermodynamics and engineering.

Applications

Heat engines, Refrigerators, Air conditioners, Steam power plants, Chemical reactions, Industrial thermodynamic processes

Advantages: 1. Measures randomness in a system. 2 Predicts whether a process is spontaneous. 3.Helps determine the maximum efficiency of engines. 4.Useful in engineering and physical chemistry.

Limitations

1. Entropy is an abstract quantity and cannot be measured directly.
2. It alone cannot determine all thermodynamic properties.
3. It does not indicate the speed of a process.

Q3. Carnot Engine.

A **Carnot engine** is an **ideal reversible heat engine** that operates between two heat reservoirs at different temperatures. It converts part of the heat absorbed from the hot reservoir into work and rejects the remaining heat to the cold reservoir.

Principle

A Carnot engine works on the **Carnot cycle**, which consists of **four reversible thermodynamic processes**:

1. Isothermal Expansion , 2. Adiabatic Expansion 3. Isothermal Compression . 4. Adiabatic Compression

Construction

A Carnot engine consists of: 1. A cylinder containing an ideal gas. 2. A frictionless movable piston. 3. A hot reservoir at temperature T_1 . 4. A cold reservoir at temperature T_2 .

Working of Carnot Cycle

1. Isothermal Expansion (A → B) 1. The gas is kept in contact with the **hot reservoir**. 2. It absorbs heat Q_1 . 3. Temperature remains constant at T_1 . 4. The gas expands and performs work.

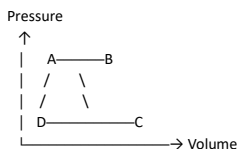
2. Adiabatic Expansion (B → C): 1. The cylinder is insulated. 2. No heat enters or leaves the system. $Q = 0$. 3. The gas continues to expand. 4. Its temperature decreases from T_1 to T_2 .

3. Isothermal Compression (C → D): 1. The gas is placed in contact with the **cold reservoir**. 2. Temperature remains constant at T_2 . 3. Heat Q_2 is rejected to the cold reservoir.

4. Adiabatic Compression (D → A): 1. The cylinder is insulated again. 2. No heat exchange takes place. 3. External work is done on the gas. 4. Temperature rises back from T_2 to T_1 . 5. The engine returns to its initial state, completing one cycle.

Diagram

Carnot Cycle (P–V Diagram)



AB → Isothermal Expansion , BC → Adiabatic Expansion , CD → Isothermal Compression , DA → Adiabatic Compression

Efficiency of Carnot Engine

Efficiency is defined as: $\eta = \frac{\text{Work Done}}{\text{Heat Supplied}}$

Since, $W = Q_1 - Q_2$

Therefore, $\eta = \frac{Q_1 - Q_2}{Q_1}$

or, $\eta = 1 - \frac{Q_2}{Q_1}$

For a reversible Carnot engine,

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$$

Hence, $\eta = 1 - \frac{T_2}{T_1}$

where, T_1 = Temperature of hot reservoir (Kelvin) , T_2 = Temperature of cold reservoir (Kelvin)

Applications: Steam power plants , Internal combustion engines , Refrigerators , Air conditioners , Thermal power stations

Advantages: 1. Maximum possible efficiency. 2. Completely reversible cycle. 3. No energy is wasted due to friction or irreversibility. 3. Acts as an ideal model for all heat engines.

Limitations: 1. It is an **ideal engine** and cannot be built perfectly in practice. 2. All processes must be **reversible**, which is not possible in real life. 3. Heat transfer at an

infinitesimally small temperature difference requires a very long time.

Q4. Black Body Radiation.

A **black body** is an ideal body that **absorbs all the incident radiation** falling on it, regardless of its wavelength or direction, and **emits the maximum possible radiation** at a given temperature.

The radiation emitted by a black body is called **black body radiation**.

Characteristics of a Black Body

1. Absorbs **100%** of the incident radiation. 2. Reflects **no** radiation. 3. Transmits **no** radiation. 4. Emits the maximum possible thermal radiation. 5. The emitted radiation depends **only on its temperature**, not on the material.

Construction of a Black Body

A practical black body is made using:

1. A hollow metal cavity. 2. A very small hole on one side. 3. The inner surface is coated with black soot.

When radiation enters through the small hole, it undergoes multiple reflections inside the cavity and is almost completely absorbed. The small hole therefore behaves like a perfect black body.

Diagram:

Black Body Radiation

When a black body is heated, it emits electromagnetic radiation called **black body radiation**.

Properties

1. It depends only on the **temperature** of the body. 2. As temperature increases: B) The amount of emitted radiation increases. A) The wavelength corresponding to maximum intensity decreases. 3. The radiation consists of a continuous spectrum.

Black Body Radiation Spectrum(graph)

Observation:

1. Higher temperature → Higher peak intensity. 2. Higher temperature → Peak shifts toward shorter wavelengths.

Important Laws of Black Body Radiation

1. Stefan–Boltzmann Law

The total energy radiated per second is proportional to the fourth power of absolute temperature.

$$E = \sigma T^4$$

where, E = Energy emitted per unit area , σ = Stefan–Boltzmann constant , T = Absolute temperature (Kelvin)

2. Wien's Displacement Law

The wavelength at which maximum radiation occurs is inversely proportional to the absolute temperature.

$\lambda_{max} T = b$ where, λ_{max} = Wavelength of maximum intensity , T = Absolute temperature , b = Wien's constant

3. Planck's Radiation Law

Planck explained black body radiation by proposing that energy is emitted in small packets called **quanta**.

The energy of each quantum is:

$E = h\nu$ where, E = Energy , h = Planck's constant ν = Frequency

This theory successfully explained the complete black body radiation spectrum.

Applications: 1. Measuring the temperature of stars. 2. Designing furnaces and industrial heaters. 3. Infrared cameras and thermal imaging. 3. Radiation thermometers (Pyrometers). 4. Space science and astrophysics.

Advantages: 1. Serves as the ideal emitter and absorber of radiation. 2. Helps explain thermal radiation. 3. Forms the basis of quantum theory. 4. Widely used in astronomy and engineering.

Limitations: 1. A perfect black body does not exist in nature. 2. It is an ideal theoretical model. 3. Practical black bodies only approximate its behavior.

Q5. Define Maxwell-Boltzmann Distribution.

The **Maxwell–Boltzmann Distribution Law** describes the **distribution of the speeds (or energies) of gas molecules** in an ideal gas at a given temperature.

It states that: "In an ideal gas, molecules move with different speeds. At a given temperature, only a few molecules have very low or very high speeds, while most molecules have an intermediate (most probable) speed."

Maxwell–Boltzmann Distribution Law

The mathematical form is:

$$f(v) = 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} v^2 e^{-mv^2/2kT}$$

where, $f(v)$ = Fraction of molecules having speed v , m = Mass of one molecule , k = Boltzmann constant , T = Absolute temperature (Kelvin) , v = Speed of the molecule

Distribution Curve (graph):

Observation: 1. At **low temperature (T_1)**: a) Curve is taller and narrower.

2. At **high temperature (T_2)**: a) Curve becomes broader and shifts to the right. 3. As temperature increases, the **most probable speed increases**.

Types of Molecular Speeds

1. Most Probable Speed (v_p)

The speed possessed by the **largest number of molecules**.

$$v_p = \sqrt{\frac{2kT}{m}}$$

2. Average Speed (v_{avg})

The arithmetic mean of all molecular speeds.

$$v_{avg} = \sqrt{\frac{8kT}{\pi m}}$$

3. Root Mean Square (RMS) Speed (v_{rms})

The square root of the average of the squares of molecular speeds.

$$v_{rms} = \sqrt{\frac{3kT}{m}}$$

Relation Between Speeds

$$v_p < v_{avg} < v_{rms}$$

This relation is **very important** and is often asked as a **1-mark or 2-mark question**.

Applications 1. Explains the motion of gas molecules. 2. Used to calculate molecular speeds. 3. Helps explain diffusion and effusion of gases. 4. Used in kinetic theory of gases. 5. Important in thermodynamics and statistical mechanics.

Advantages: 1. Explains the distribution of molecular speeds. 2. Supports the kinetic theory of gases. 3. Predicts average, RMS, and most probable speeds. 4. Helps understand gas behavior at different temperatures.

Limitations: 1. Applicable only to ideal gases. 2. Fails at very low temperatures. 3. Does not explain the behavior of bosons or fermions. 4. Not valid for quantum particles.

Conclusion

The Maxwell–Boltzmann Distribution explains how the speeds of gas molecules are distributed in an ideal gas. It shows that molecules have different speeds and that most molecules move with the most probable speed. As temperature increases, molecular speeds also increase. The distribution is fundamental to the kinetic theory of gases and statistical mechanics.

Q6.Define Maxwell Relations.

Maxwell Relations are a set of four thermodynamic equations derived from the thermodynamic potentials. They establish relationships between measurable thermodynamic quantities such as pressure (P), volume (V), temperature (T), and entropy (S).

Thermodynamic Potentials

The four thermodynamic potentials are:Internal Energy (U) ,Enthalpy (H) ,Helmholtz Free Energy (F or A) ,Gibbs Free Energy (G)

These potentials are used to derive the four Maxwell relations.

1.First Maxwell Relation

From Internal Energy (U), $dU = TdS - PdV$

Therefore, $\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$

2.Second Maxwell Relation

From Enthalpy (H), $dH = TdS + VdP$

Therefore, $\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$

3.Third Maxwell Relation

From Helmholtz Free Energy (F or A), $dF = -SdT - PdV$

Therefore, $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$

4. Fourth Maxwell Relation

From Gibbs Free Energy (G), $dG = -SdT + VdP$

Therefore, $\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$

Diagram (Thermodynamic Potentials)

Thermodynamic Potentials

Internal Energy (U)

$$dU = TdS - PdV$$

First Maxwell Relation

Enthalpy (H)

Helmholtz Free Energy (F)

$$dH = TdS + VdP$$

$$dF = -SdT - PdV$$

Second Maxwell Relation

Third Maxwell Relation

Gibbs Free Energy (G)

$$dG = -SdT + VdP$$

Fourth Maxwell Relation

Applications: 1.Calculation of entropy changes. 2. Study of heat engines and refrigerators. 3.Chemical thermodynamics. 4.Statistical mechanics. 4.Engineering and power plant analysis.

Advantages: 1.Reduce complex calculations. 2.Relate measurable properties such as pressure, volume, and temperature. 3.Useful in solving thermodynamic problems. 4.Form the foundation of advanced thermodynamics.

Limitations 1.Applicable only to systems in **thermodynamic equilibrium**. 2.Not directly applicable to irreversible processes. 3Require thermodynamic potentials for derivation.

Conclusion

The **Maxwell Relations** are four important equations obtained from the thermodynamic potentials. They connect pressure, volume, temperature, and entropy through partial derivatives, making many thermodynamic calculations easier. These relations are fundamental in the study of thermodynamics and have wide applications in physics, chemistry, and engineering.

